GROGO□I.MARKET

Master Control Document

Gap Resolution, Alignment & Authority Reference

Document ID	GROGONI-MCD-001
Version	1.0
Status	Active — Supersedes conflicting content in Docs 1–3
Authority	Where this document conflicts with Docs 1, 2, or 3, this document takes precedence
Covers	Doc 1 (Architecture v3), Doc 2 (Technical Spec v1), Doc 3 (PRD v1)
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1. Purpose & Authority

This Master Control Document (MCD) exists to do three things:

- Fill gaps — add the decisions that are missing from the three companion documents.
- Resolve conflicts — where Doc 2 and Doc 3 contradict each other, this document rules.
- Set authority — any team member uncertain which document to follow should default to this one.

This document does not replace the companion documents. It sits above them. Read the companion docs for detail; come here to resolve disagreement or find what is absent.

1.1 Document Hierarchy

Priority	Document	Role
1 (Highest)	This MCD — GROGONI-MCD-001	Gap fills, conflict resolution, final authority
2	Doc 2 — Technical Spec v1.0	Engineering implementation detail
3	Doc 3 — PRD v1.0	Product scope, user stories, delivery roadmap
4	Doc 1 — Architecture v3.0	Vision, data model, system behaviour

2. Cross-Document Conflict Register

The following conflicts were identified between the companion documents. The ruling in the rightmost column is binding.

ID	Topic	Conflict	Ruling (this document governs)
C-01	**Sprint roadmap vs Milestones**	Doc 3 defines 10 sprints. Doc 2 defines 6 milestones. They do not map to each other.	The sprint map in Doc 3 is the delivery unit. The milestones in Doc 2 are release gates. Section 3 of this document provides the unified mapping. Teams use sprints for day-to-day delivery and milestones for release decisions.
C-02	**AI assistant timing**	Doc 3 places AI assistant in Sprint 5. Doc 2 Milestone 5 is also AI Runtime, but after 4 infrastructure milestones.	AI must enter in Sprint 1 as a thin command parser and grow each sprint. Full AI Runtime remains Milestone 5 for cloud/model features, but a basic local parser is Sprint 1 non-negotiable. See Section 4.
C-03	**Definition of Done**	Doc 3 has a Definition of Done checklist. Doc 2 has performance targets and testing strategy. Neither references the other.	The single Definition of Done is defined in Section 5 of this document. It incorporates both sources. Doc 3 Section 12 and Doc 2 Section 25 are superseded by Section 5 here.
C-04	**Marketplace scope**	Doc 3 includes Marketplace as Epic 8 and Sprint 10 (in scope). Doc 1 Phase 9 treats it as a late phase.	Marketplace is out of scope for MVP and Beta. It enters the roadmap only after Public Launch stability is confirmed. Sprint 10 in Doc 3 is deferred. See Section 6.
C-05	**Offline sync conflict rules**	Doc 1 Section 8.3 lists 4 rules. Doc 2 Section 9.4 lists 5 rules with slightly different wording on the 'latest action wins' condition.	The authoritative conflict rules are defined in Section 7 of this document. Both prior versions are superseded.
C-06	**Performance targets**	Doc 3 states chat response < 2s. Doc 2 states basic local action < 500ms and chat screen load < 3s. These are measuring different things without clear definitions.	Unified performance targets are defined in Section 8 of this document with clear measurement definitions.
C-07	**Payment providers**	Doc 1 lists M-Pesa, Stripe, PayPal, Pogo, MoneySwitch. Doc 2 Section 18.1 lists M-Pesa, Stripe, PayPal only as initial, with regional providers later.	Doc 2 is correct. Initial: M-Pesa and manual recording only. Stripe and PayPal are Milestone 3 stretch. Pogo and MoneySwitch are post-launch regional expansion.
C-08	**Team structure**	Doc 3 Section 16 lists 10 core roles. No team size or timeline is attached to any sprint or milestone in any document.	The binding team model and sprint duration assumptions are defined in Section 9 of this document.

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3. Unified Sprint-to-Milestone Map

This is the single authoritative delivery plan. Sprint duration: 2 weeks. Team size assumption: 5 engineers (1 mobile, 1 backend, 1 AI/ML, 1 full-stack, 1 QA). Total calendar time to Public Launch: approximately 6 months. See Section 9 for team model.

Sprint	Deliverables	Milestone Gate	Release Stage	Exit Condition
S1	OTP auth, business creation, language selection, basic rule-based chat parser (local, no model)	**M1 begins**	Internal Alpha	Owner can log in and type a command that is understood
S2	Products CRUD, customers CRUD, AI parser handles product + customer intents with confidence scoring	—	Internal Alpha	Agent correctly handles 10 predefined commands with >80% accuracy on test set
S3	Invoice creation (chat-driven), inventory tracking, tax line calculation	**M1 complete**	Internal Alpha	End-to-end: chat → invoice → PDF in under 2 seconds on low-end device
S4	Local IndexedDB, offline queue, delta sync, conflict resolution, retry logic	**M2 complete**	Internal Alpha	Create invoice offline, reconnect, sync succeeds with no data loss
S5	Manual payments, M-Pesa webhook integration, payment status on invoice, debt tracking	**M3 begins**	Closed Beta	Payment received → invoice marked paid within 30 seconds of webhook
S6	Spreadsheet and PDF import, field extraction, preview and confirmation flow	**M3 / M4 complete**	Closed Beta	Upload a supplier CSV → products appear in confirmation screen → saved correctly
S7	Full AI Runtime: Intent Router, Planner, Tool Executor, memory, context builder, cloud fallback	**M5 complete**	Closed Beta	AI task completion rate >85% on 50-command evaluation set
S8	Reports (sales summary, debt report, stock report), audit logs, notification channels	—	Closed Beta	Owner can generate a monthly sales report from chat
S9	Logistics module, vehicle and driver records, delivery tracking	**M6 complete**	Public Beta	Delivery dispatched → customer notified → status updated on completion

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Sprint	Deliverables	Milestone Gate	Release Stage	Exit Condition
S10	Performance hardening, accessibility, regional tax config, SMS fallback, USSD stub	**Public Launch**	Public Launch	All NFRs from Section 8 pass on 1 GB RAM test device

4. AI Implementation Strategy (Gap Fill)

The companion documents describe AI goals and architecture but omit three critical operational decisions: when AI enters the build, what accuracy is required before each release, and what the product looks and feels like when the AI fails. This section fills those gaps.

4.1 AI Entry Point

AI enters in Sprint 1, not Sprint 5. The Sprint 1 AI is a deterministic rule-based command parser — no model, no inference, no cloud. It maps typed phrases to intents using pattern matching. This is intentional: it forces the team to design the intent taxonomy, confirmation flows, and error states before any model is involved. Every later AI upgrade is a drop-in replacement behind the same interface.

Sprint	AI Mode	Capability	Accuracy Gate
1–2	**Rule-based parser**	Keyword → intent mapping. Handles ~20 predefined command patterns.	>80% on internal 50-command test set before Sprint 3 begins
3–4	**Rule-based + entity extraction**	Extracts product names, quantities, customer names from natural phrasing.	>85% entity extraction accuracy on 100-phrase eval set
5–6	**Tiny local SLM (optional, Wi-Fi download)**	Handles ambiguous phrasing, mixed-language input, clarification questions.	>85% intent accuracy on 200-command multilingual eval set
7	**Full AI Runtime (cloud fallback)**	Complex reasoning, document extraction, long-context tasks.	>90% task completion rate; cloud fallback rate <30% of all requests

4.2 AI Failure Protocol

This is the most important gap in the companion documents. The product must define exactly what happens at each confidence level. These rules are binding across all AI modes:

Confidence	System Behaviour	User Experience
>0.85	Execute with confirmation for medium/high risk actions. Execute silently for read-only.	Agent says what it understood and what it did. One-tap undo available for 30 seconds.
0.60–0.85	Ask one clarifying question before acting. Do not guess.	Agent says: 'Did you mean [most likely intent]?' with 2–3 tap options.
0.40–0.60	Do not attempt execution. Show shortcut menu of common actions.	Agent says: 'I am not sure what you meant. Here are common actions:' + 4 quick-tap buttons.

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Confidence	System Behaviour	User Experience
<0.40	Do not attempt execution. Log the unrecognised phrase for review.	Agent says: 'I did not understand that. Try saying something like: Add 10 bags of maize.'
3 consecutive failures	Offer to open a structured form for the last attempted action.	Agent says: 'Would you like me to open a form instead?' Two-tap: Yes / No.

4.3 AI Evaluation Dataset Requirements

Before Sprint 3 begins, the team must produce and maintain an evaluation dataset. This dataset is the source of truth for all accuracy gates above.

- Minimum 200 commands at Sprint 3. Minimum 500 by Sprint 7.
- At least 40% of commands must be in Swahili or Swahili-English mixed language.
- Must include deliberately ambiguous phrases to test clarification behaviour.
- Must include adversarial inputs (gibberish, very long messages, repeated commands).
- Dataset is version-controlled in the monorepo under /tests/ai-eval/.
- Evaluation runs automatically on every PR that touches the AI runtime.

5. Unified Definition of Done

This supersedes Doc 3 Section 12. A feature is not complete until every item below is checked. No exceptions without a written waiver from the Engineering Lead.

5.1 Code Quality

- Code implemented and peer-reviewed (minimum 1 reviewer who did not write the code).
- No new lint errors or type errors introduced.
- No TODO comments left in production code paths.

5.2 Testing

- Unit tests pass with >80% coverage on new business logic.
- Integration tests pass for all API routes introduced.
- Offline test: feature works with network disabled and syncs correctly on reconnect.
- Device test: feature verified on a physical or emulated 1 GB RAM Android device.
- If the feature touches AI: AI evaluation suite run and accuracy gate met (see Section 4.1).

5.3 Safety & Security

- AI safety review: no new high/critical risk tool can execute without a confirmation step.
- RBAC check: feature respects role permissions. Owner-only features are gated.
- Sensitive data is not logged in plaintext (payment references, phone numbers).

5.4 Performance

- Chat response for local actions: under 2 seconds on low-end device (see Section 8).
- No new API route with p95 latency above 500ms under 50 concurrent users.
- App shell size has not increased by more than 200 KB for this feature.

5.5 Product

- All acceptance criteria in the user story are met.
- Feature is reachable from the chat interface without navigating a menu.

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- Feature works in English and Swahili (text labels, AI commands, error messages).
- Accessibility: interactive elements meet 44x44 pt minimum tap target.

5.6 Documentation

- API routes documented in /docs/api/ with request/response examples.
- If new tool defined: tool definition JSON added to tool registry.
- CHANGELOG entry added.

6. Marketplace Scope Ruling

Conflict C-04 ruled that the Marketplace (Doc 3 Epic 8 / Sprint 10) is deferred. This section defines what that means precisely.

6.1 What Is Deferred

- Skill installation, skill registry, skill sandbox.
- Third-party developer access, skill signing and verification.
- Skill marketplace listing, reviews, updates, revenue sharing.
- Industry template packs sold through the marketplace.

6.2 What Remains In Scope

- Internal first-party skills: the team may build and ship pre-installed skills (e.g. a KRA tax calculator) that ship as part of the app. These are not marketplace items — they are features.
- The skill architecture in Doc 2 Section 21 should still be designed and partially built, so that marketplace can be activated post-launch without a rewrite.
- The tool permission model, sandbox interface, and skill.json spec should be finalised in the technical design during Sprint 7.

6.3 Trigger Conditions for Reactivation

The marketplace roadmap is reactivated when all three conditions are met after Public Launch:

- 500+ active businesses on the platform for 60+ consecutive days.
- Sync success rate >98% for 30 consecutive days.
- At least 3 identified external developers who have requested API access.

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7. Authoritative Sync Conflict Rules

Conflict C-05: Doc 1 Section 8.3 and Doc 2 Section 9.4 contain inconsistent conflict resolution rules. The following rules supersede both. These rules are implemented in the Sync Service and are not overridable by client code.

Rule	Condition	Action
SR-1	Any conflict involving a payment amount, transaction reference, or payment status	BLOCK. Do not auto-resolve. Surface to account owner with both versions. No write until owner selects one. Log both versions permanently.
SR-2	Any conflict involving invoice total, tax amount, or discount amount	BLOCK. Do not auto-resolve. Surface to the user who created the invoice. Preserve both versions in the event log.
SR-3	Any conflict involving product quantity	BLOCK. Surface to owner with both values and the timestamps of each action. Owner selects correct value.
SR-4	Non-financial fields (product name, customer address, notes) where two edits conflict	Apply the edit with the later server-confirmed timestamp. Log the overwritten value in the event log for 90 days.
SR-5	Two records created offline that appear to be duplicates (same customer name + phone, or same invoice number)	Do not merge. Create both. Flag both as potential duplicates in the UI. Owner resolves manually.
SR-6	A deletion conflicts with an edit (one device deleted a record; another device edited it)	Prefer the edit. Restore the record. Flag it as 'deletion attempted' in the audit log. Notify the owner.
SR-7	Any conflict where the correct resolution cannot be determined by rules SR-1 through SR-6	Preserve both versions. Mark the record as 'conflict — pending resolution'. Surface in a dedicated conflict resolution screen. Never silently discard either version.

8. Unified Performance Targets

Conflict C-06: performance targets across documents measured different things with the same names. The following definitions and targets are authoritative. All targets are measured on the reference device: a 1 GB RAM Android phone on a 3G connection unless stated otherwise.

Metric	Definition	Target	Measured At
App shell load (cold start)	Time from tap to chat input visible, no data loaded	**< 3s**	Device, offline cache warm
Local action response	Time from user send to agent reply for fully local actions (e.g. 'show my products')	**< 500ms**	Device, offline
Invoice creation (chat, online)	Time from user sends final confirmation to invoice saved and shown	**< 2s**	Device, 3G
Invoice creation (chat, offline)	Time from user sends final confirmation to invoice saved locally	**< 1s**	Device, offline
Sync cycle (small delta)	Time to push 10 events and receive server response	**< 30s**	3G connection
AI cloud fallback response	Time from message sent to response received using cloud model	**< 4s**	3G connection
Document import (spreadsheet)	Time from upload complete to preview shown (async job)	**< 60s**	Server-side, measured from upload ACK
API p95 latency (core routes)	95th percentile response time for /products, /customers, /invoices, /payments	**< 500ms**	Server under 50 concurrent users
Payment webhook processing	Time from webhook received to invoice status updated	**< 30s**	Server
Crash-free session rate	Percentage of sessions with no JS exception or app crash	**> 99.5%**	Production telemetry

9. Team Model & Delivery Assumptions

Conflict C-08: no document attaches a team size or calendar timeline to the sprint roadmap. This section fills that gap. All timeline estimates in this document assume the following.

9.1 Core Team (MVP to Public Launch)

Role	Responsibility in Grogoni	Enters At
Product Manager / Owner	Sprint roadmap, user stories, stakeholder communication, acceptance sign-off	Sprint 1
Technical Architect	System design, PR review, technical decision authority	Sprint 1
Mobile / Frontend Engineer	PWA, chat UI, offline DB, sync queue, camera, contacts	Sprint 1
Backend Engineer	API service, sync service, database, auth, webhooks	Sprint 1
AI / ML Engineer	Intent router, AI runtime, eval dataset, model integration, cloud fallback	Sprint 1
QA Engineer	Test plans, device testing, AI eval runs, offline testing, security review	Sprint 2
UX Designer	Chat UI patterns, form designs, onboarding flow, error states	Sprint 1 (part-time), Sprint 3 (full-time)
DevOps Engineer	Docker, CI/CD, cloud infra, observability, secrets management	Sprint 2

9.2 Calendar Timeline

Phase	Sprints	Calendar Duration	Gate
Internal Alpha	S1–S4	8 weeks (~2 months)	Internal team can replace paper ledger with the app
Closed Beta	S5–S7	6 weeks (~1.5 months)	10 selected merchants actively using the app daily
Public Beta	S8–S9	4 weeks (~1 month)	50 businesses onboarded; sync and payment stable

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Phase	Sprints	Calendar Duration	Gate
Public Launch	S10	2 weeks	All NFRs from Section 8 pass; support process in place

10. Remaining Gap Fills

The items below are decisions required to build the product that do not appear in any companion document. Each is authoritative.

10.1 Cloud AI Cost Control

No companion document addresses the cost of cloud AI fallback at scale. The following controls are mandatory from Sprint 7 onwards:

- Per-account daily cloud AI call limit: 200 calls/day on the free tier; 1,000/day on Business; unlimited on Pro.
- If a user hits their daily limit, the AI downgrades to rule-based mode and notifies the user.
- Cloud fallback is never triggered for actions the rule-based parser can handle with >0.85 confidence.
- Cloud AI spend is tracked per account and surfaced in the admin dashboard.
- A monthly cloud AI budget alert fires when the platform-wide cost exceeds a pre-set threshold.

10.2 Agent Action Rate Limits

No companion document addresses what happens if an AI agent issues a rapid sequence of harmful tool calls (hallucination or injection attack). The following limits are enforced by the Tool Executor:

- Maximum 10 tool executions per conversation turn. Any turn that would require more is blocked and surfaced as an error.
- Maximum 50 high-risk or critical tool calls per account per hour. Exceeded limit triggers owner notification and temporary agent suspension.
- No tool can be called more than 5 times in a single conversation turn (duplicate call detection).
- Any tool call that would affect more than 20 records at once requires explicit owner confirmation, regardless of risk level.

10.3 Onboarding Success Definition

Doc 3 Section 5 states 'phone onboarding under 3 minutes' and 'first invoice within 5 minutes' as launch goals but provides no funnel definition. The binding onboarding funnel is:

Step	Action	Target Time	Drop-off Threshold
1	Phone number entry + OTP verification	< 60s	Alert if >15% of users drop here
2	Business name + type selection (3 taps)	< 30s	Alert if >10% drop here
3	Language selection	< 15s	Alert if >5% drop here
4	Agent introduction + first chat message sent by user	< 60s	Alert if >20% drop here
5	First product added (via chat)	< 90s	Alert if >30% drop here

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Step	Action	Target Time	Drop-off Threshold
6	First invoice created	< 3 min from step 5	Alert if >40% drop here

10.4 Minimum Viable Swahili Support

No companion document specifies what 'language support' means concretely. The following is the minimum required before Closed Beta:

- All UI labels, button text, error messages, and agent responses available in Swahili.
- AI command parser trained on 40%+ Swahili and Swahili-English mixed commands.
- Number and date parsing handles Swahili terms: leo (today), jana (yesterday), kesho (tomorrow).
- Currency input accepts both '3,500' and '3500' and 'sh 3500' and 'KES 3500'.
- Agent responses do not mix languages within a single message unless the user's input was mixed.

10.5 Data Deletion & Export (GDPR-equivalent)

No companion document specifies the technical implementation of data rights. The following is binding:

- Account deletion request: owner can submit from Settings. Account is deactivated immediately. All PII (name, phone, email) is anonymised within 30 days.
- Business records (invoices, payments, tax records) are retained for 7 years in anonymised form for legal compliance.
- Data export: owner can request a full export from Settings. Export is prepared as a ZIP (JSON + PDF invoices) and delivered via download link within 24 hours.
- Staff member removal: staff data is anonymised on removal but their actions remain in the audit log attributed to 'Former Staff Member'.

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11. Decision Log

This section records decisions made in this document that are not derivable from the companion documents. Any future amendment to these decisions must be recorded here with a version increment.

ID	Decision	Rationale
D-01	AI parser enters Sprint 1 as a rule-based system, not Sprint 5 as a full model	Product promise is chat-first. Delaying AI to Sprint 5 means building everything else without validating the core UX.
D-02	Marketplace is deferred until post-launch trigger conditions are met	Building a marketplace before product-market fit risks wasting 4–6 weeks of engineering on infrastructure that may need a complete redesign once real developers use it.
D-03	Sprint is the primary delivery unit; milestones are release gates	Two teams were at risk of using different planning cadences. Sprints are operational; milestones are contractual/release.
D-04	Cloud AI fallback rate limit: 200 calls/day free tier	Without a limit, a single power user could generate a cost that exceeds their subscription value within days.
D-05	Sync Rule SR-6 prefers edits over deletions	Accidental deletion of a record that another device has edited is a more disruptive data loss than preserving a deleted record. The audit log captures the deletion intent.
D-06	Payment providers at launch: M-Pesa and manual recording only	Stripe and PayPal integration timelines are dependent on KYC tier completion and entity registration. Shipping without them is lower risk than delaying launch.
D-07	Agent action budget: max 10 tool calls per turn	Prevents runaway AI loops from exhausting database write capacity or sending unintended bulk messages.

End of Grogoni Master Control Document v1.0

This document supersedes conflicting content in companion documents. Questions: raise with the Technical Architect.